
Emma Zhou

Bilingual Agent Platform Intern

Emma Zhou | Bilingual Agent Platform Intern | emma.zhou.eval@example.com | Shanghai / Remote | Available for 6 months. Focused on grounded Agent workflows, durable execution, multilingual retrieval and observable tool use. Comfortable turning ambiguous user goals into typed plans while keeping evidence, permissions and completion criteria explicit.

EDUCATION

Tongji University, M.Eng. in Software Engineering, 2025.09-2028.06, GPA 3.86/4.00. Research topics include LLM application engineering, information retrieval and reliable distributed workflows. South China University of Technology, B.Eng. in Computer Science, 2021.09-2025.06, outstanding graduate. Coursework: databases, distributed systems, machine learning, natural language processing, software testing and human-computer interaction.

TECHNICAL SKILLS

Languages: Python, TypeScript and SQL. Agent: LangGraph state graphs, tool contracts, checkpoint recovery, bounded retries, human approval and completion gates. Retrieval: RAG, multilingual embeddings, BM25, exact match, multi-query, RRF, reranking and citation verification. Engineering: FastAPI, Redis, PostgreSQL, Docker, pytest and OpenTelemetry. Evaluation: Recall@K, MRR, nDCG, task success, grounding rate, latency percentiles and provider token accounting.

INTERNSHIP EXPERIENCE

AI Platform Intern, Mock Cloud Lab, 2026.03-2026.08. Delivered a Chinese-English research assistant with citation verification. Built a bilingual tool-result grounding pipeline. Added typed tool schemas, idempotency keys, bounded exponential backoff and trace dashboards; collaborated with product and QA to convert twelve failure reports into regression cases. Separated transient network retries from semantic repair, and prevented outbound side effects from being replayed after uncertain responses.

Research Engineering Intern, Mock Knowledge Systems Group, 2025.07-2025.12. Implemented document ingestion, metadata filters and citation-level evaluation for bilingual technical reports. Diagnosed two-column reading-order failures and mixed text-scan PDFs, preserving page provenance through extraction and chunking. Built a review queue for unsupported claims and helped annotate hard negatives that share technical vocabulary but answer different user intents.

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RESEARCH

Studied query rewriting for mixed Chinese-English terminology. Compared single-query, multi-query and HyDE-style retrieval under the same candidate pool; reported Recall@5, MRR, nDCG and latency instead of relying on answer fluency. Designed ablations for query source, chunk boundaries, reranker input and evidence thresholds. Distinguished a relevant but insufficient passage from a contradictory passage, and required every generated conclusion to map back to a source span.

PROJECTS

Bilingual Research Assistant: Reduced unsupported claims from 11.8% to 2.1%. The system preserves source spans and rejects answers whose citations cannot be found verbatim. It rewrites long requests into complementary retrieval views, merges candidates with reciprocal-rank fusion and records which query retrieved each passage.

Agent Replay Console: reconstructed state transitions and tool receipts from checkpoints, while separating business artifacts from conversational memory. Added stale-run recovery, checkpoint history inspection and idempotent artifact reconciliation after process crashes.

Evaluation Harness: created adversarial cases for prompt injection, early stopping, repeated tool calls, malformed JSON, weak evidence and cross-user context leakage. Reports stage-level failures, pass^k, latency, token cost and regression provenance rather than a single opaque score.

Document Intake Pipeline: routed native text pages and scanned pages independently, detected repeated headers, retained page numbers and produced cross-page context only when a sentence or bullet genuinely continued.

LEADERSHIP & AWARDS

Maintained an open-source evaluation toolkit, reviewed pull requests and wrote reproducible failure reports. Organized a twelve-week Agent engineering reading group covering retrieval, tool governance, durable execution and evaluation. Received a university innovation scholarship and a regional software design award. English CET-6; comfortable presenting technical design reviews in Chinese and English and writing bilingual engineering documentation.